

The 3D-Slicer auto-Segmentation instruction and the program

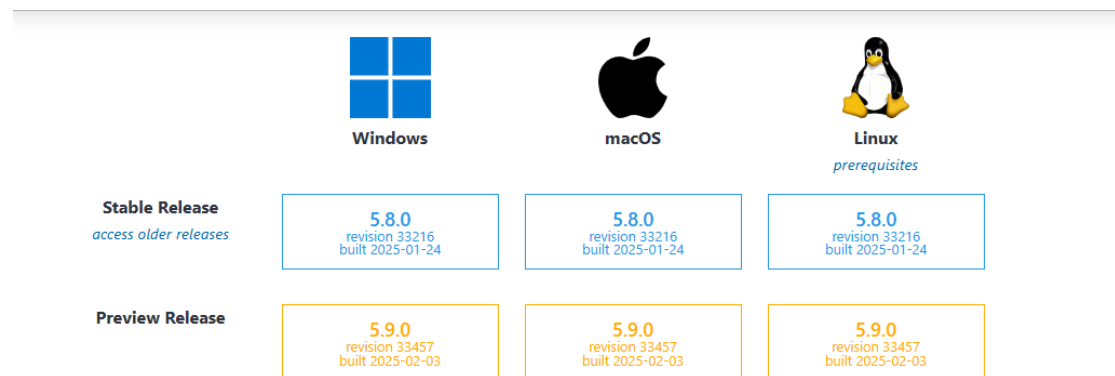
Mingyu Shang

University College London

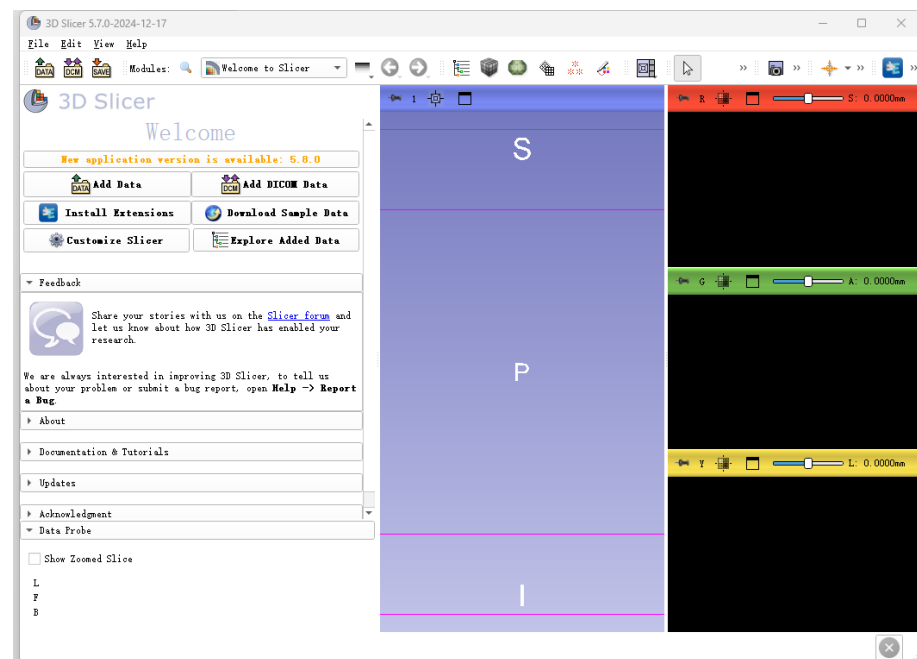
email: zcemms6@ucl.ac.uk

Install the 3D-Slicer and the modules:

[Download 3D Slicer | 3D Slicer](#)



Make sure you download 5.9.0



Select "extension"



Install Extensions



DentalSegmentator

Segmentation

Fully automatic AI segmentation tool for Dental CT and CBCT scans based on DentalSegmentator nnU-Net model.



UNINSTALL

Install “DentalSegmentator”

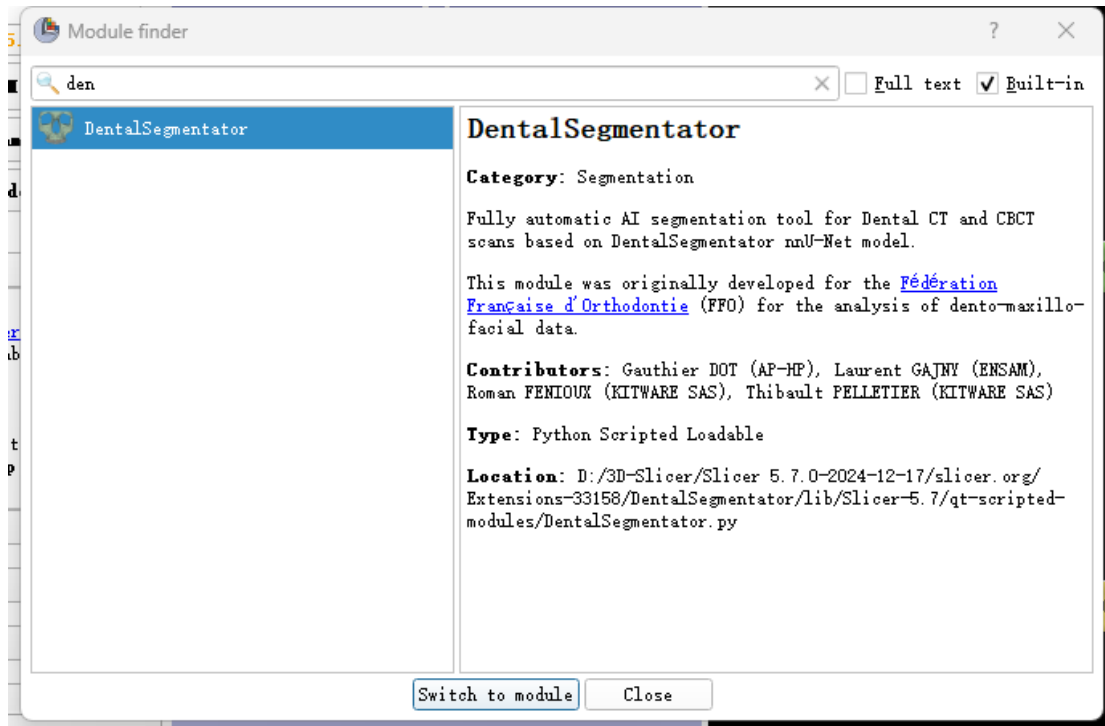
Back to the main pages and select “find modules” on the top

Modules: 

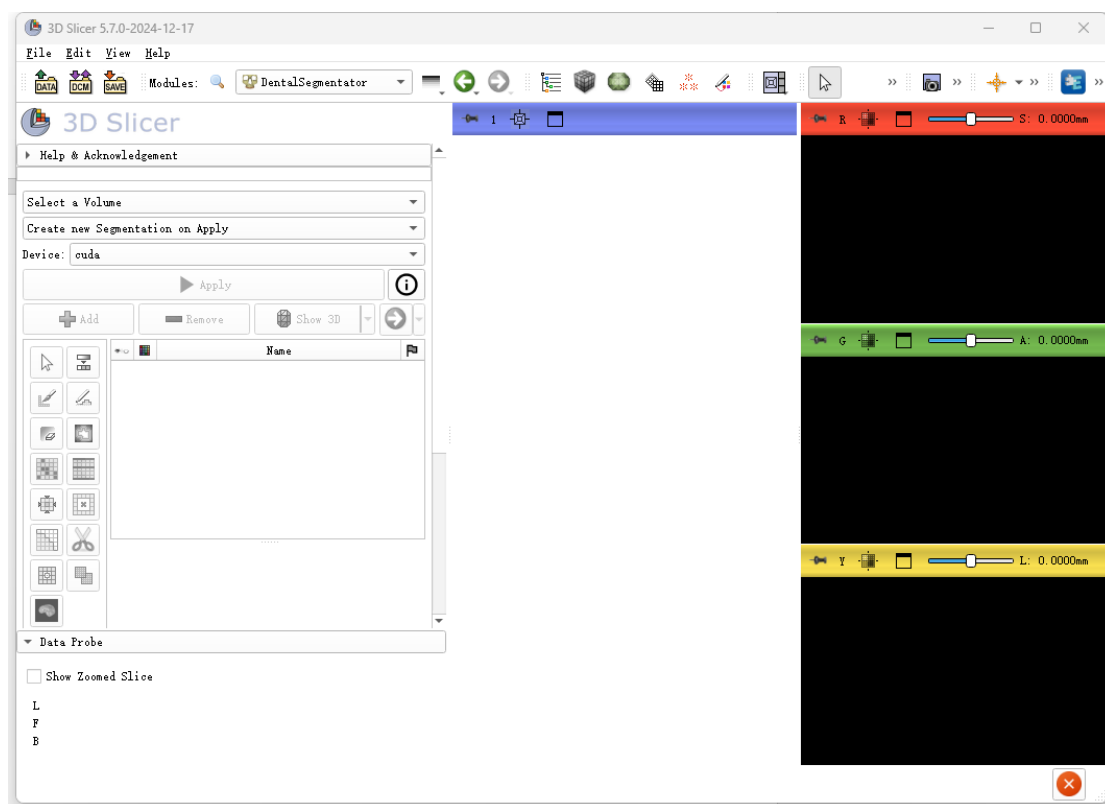


Welcome to Slicer





Find “DentalSegmentator” and switch to module



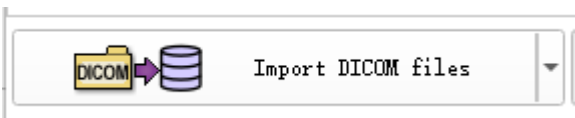
We have installed 3D-Slicer and DentalSegmentator.

Import the data:

Back to the main page. We can see



We Select “add DICOM Data”



Then “Import DICOM files”, we can import our data to the 3D-Slicer

Patient name	Patient ID	Birth date	Sex	Studies	Last study date	Date added	
CLEMENTS, Lexi	21803920	2010-08-25	F	1	2024-09-26	2024-1...33.558	
Christine							
JillianMISS							
REINECKE,							
Matthew	21664824	2013-02-23	M	1	2024-09-04	2024-1...31.108	
GeorgeMR.							
ARORA, Daksh	21471256	2010-06-07	M	1	2024-08-22	2024-1...26.467	
Master							
DOCKERY,							
Makeda Rickita	21627944	2008-02-08	F	1	2024-06-27	2024-1...23.943	
KarleneMiss							
GREEN, Charles	21433173	2013-03-28	M	1	2024-03-14	2024-1...21.362	
FrancisMaster							

The data can be shown, then we can select the data we want.

Use DentalSegmentator:

After we select our data, we change the module to

DentalSegmentator

001001002: Unnamed Series

Create new Segmentation on Apply

Device: cuda

Apply

+

Add

Remove

Show 3D

↩

ⓘ

Just Select “Apply”. The 3D model would be formed.

Notice for the first time, it can not work directly because the lack of packages. You may need to type these codes in the python console



Thanks for the guidance! After carefully reading and examining what you directed me to, I entered the following lines into the python console.

```
pip_install("TQDM")
pip_install("blosc2")
pip_install("nibabel")
pip_install("dynamic_network_architectures")
pip_install("batchgeneratorsv2")
pip_install("matplotlib")
pip_install("Seaborn")
```

After a few days of being nervous. I didn't believe I would see this program work, but now everything works perfectly!

pip_install("tqdm")

pip_install("blosc2")

pip_install("nibabel")

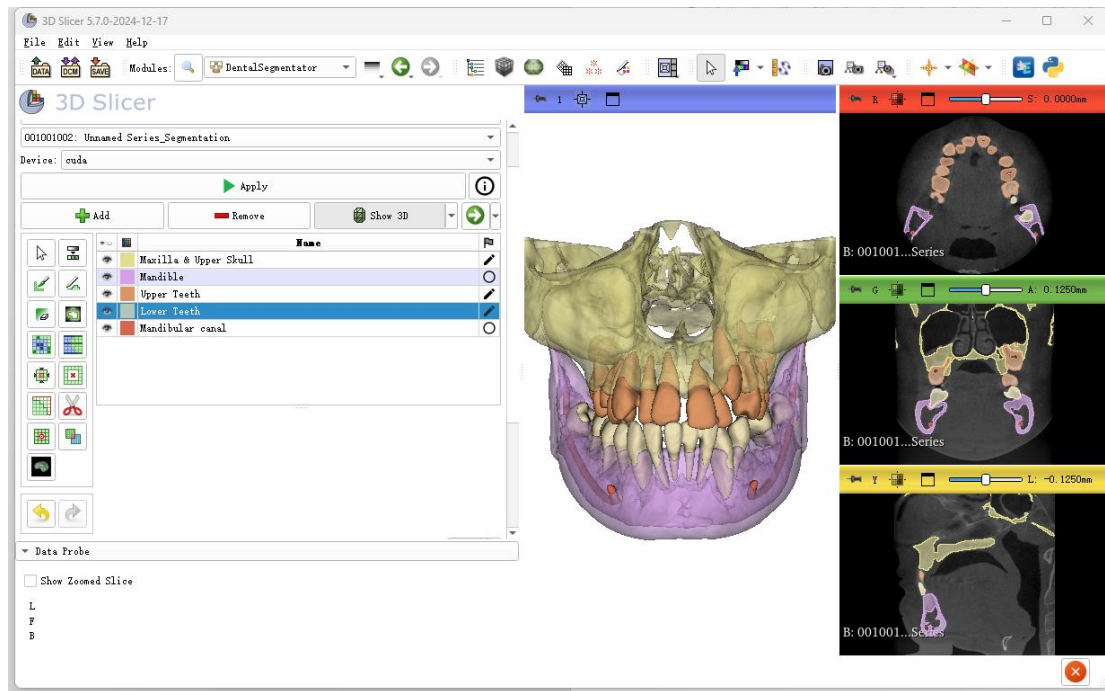
pip_install("dynamic_network_architectures")

pip_install("batchgeneratorsv2")

pip_install("matplotlib")

pip_install("seaborn")

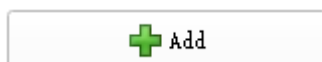
I put the code here to copy.



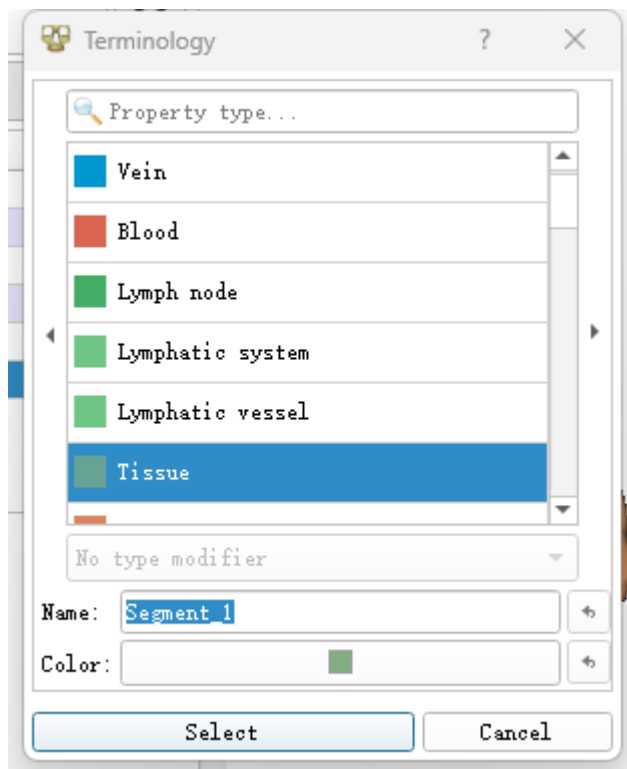
A 3D model is made!!!

The segmentation for particular tooth:

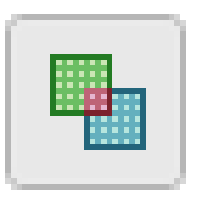
For example, we want one of upper incisor.



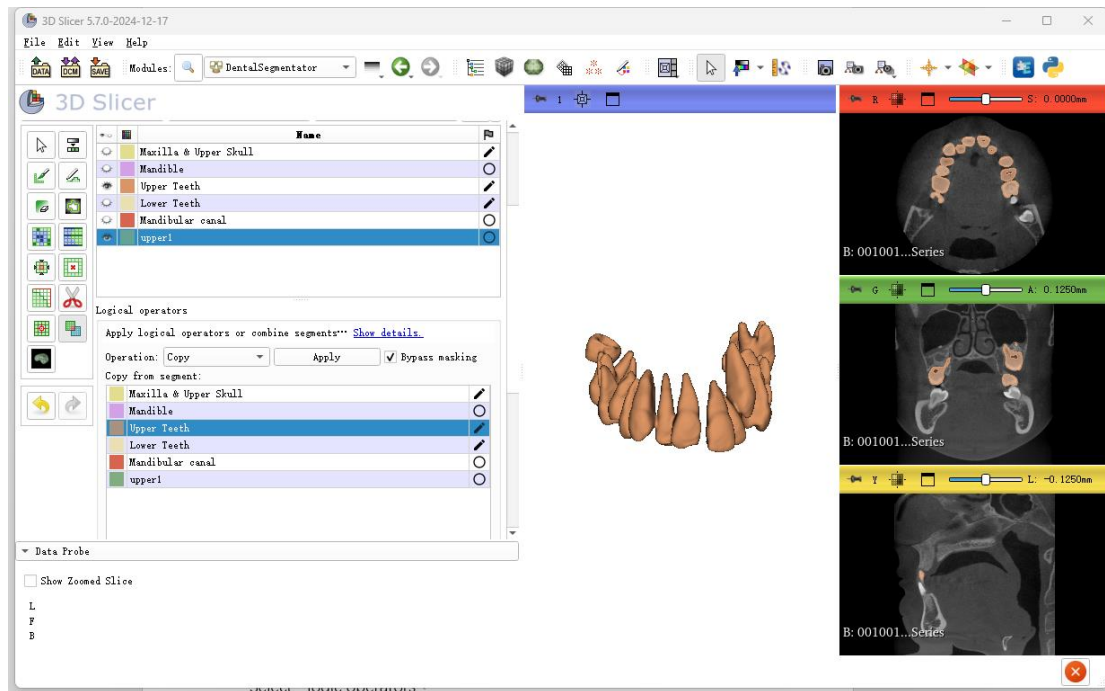
Select “add” to add a new segmentation



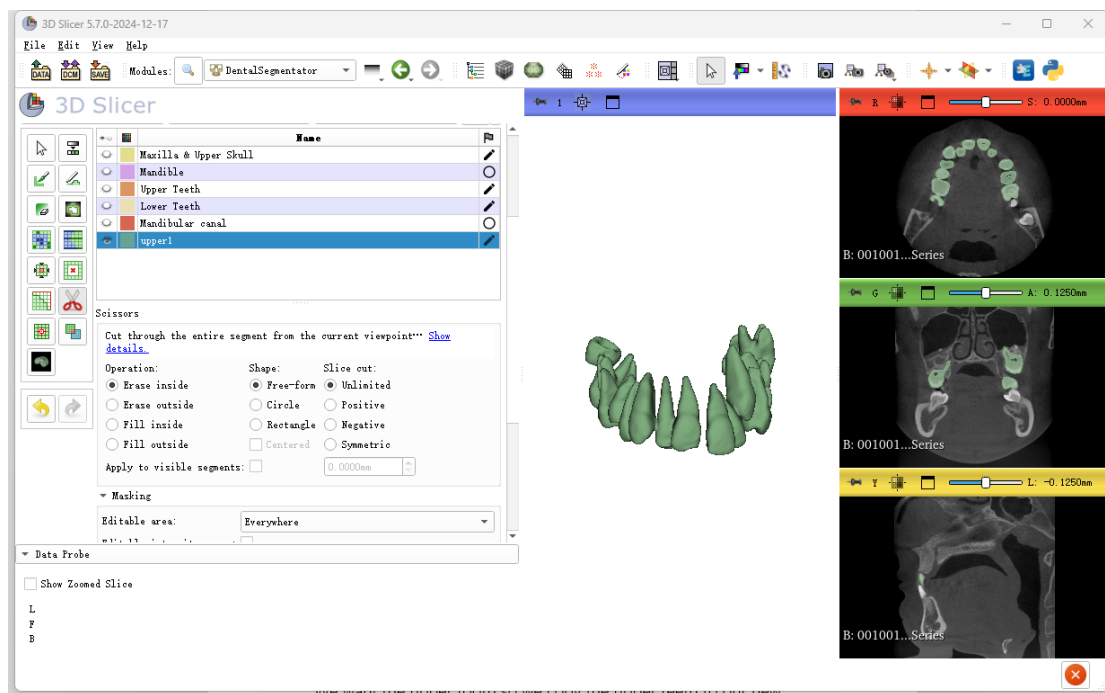
Rename the new segmentation.



Select “ logic operators”



We want the upper tooth so we copy the upper teeth to our new segmentation. Operation: copy the select “apply”



Now we choose our new segmentation and use scissors

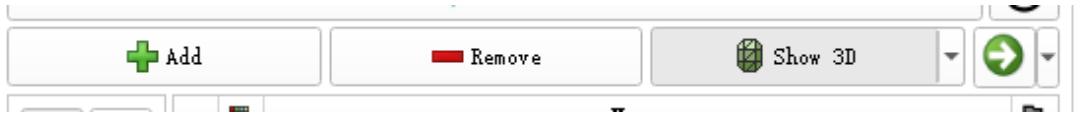
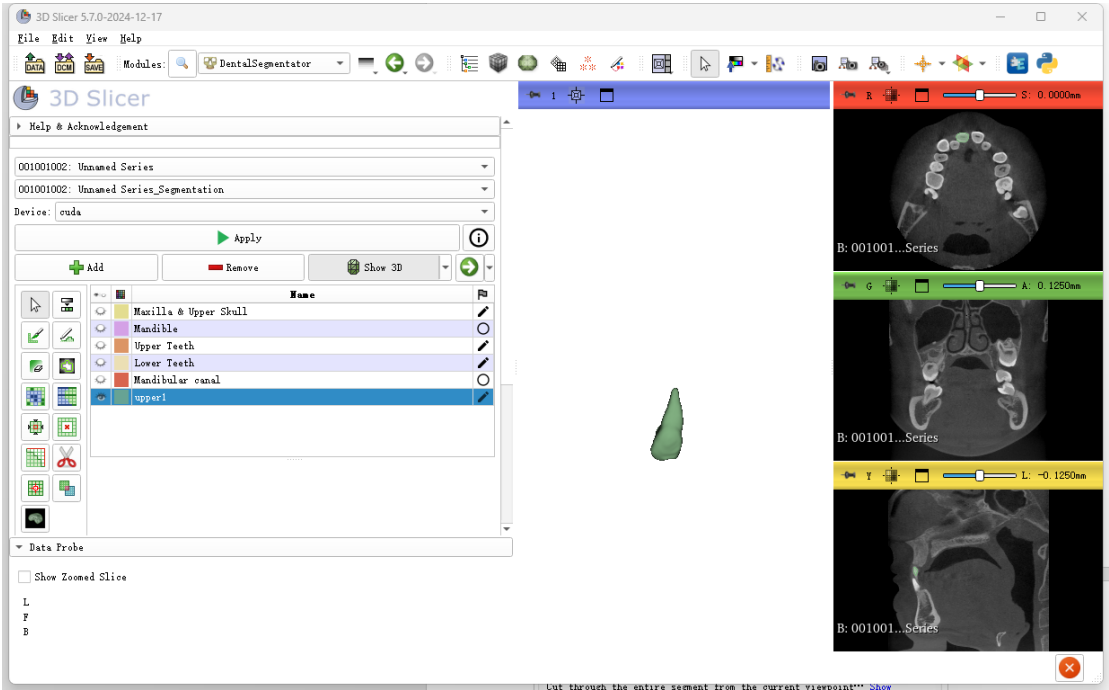
Cut through the entire segment from the current viewpoint... [Show details.](#)

Operation:	Shape:	Slice cut:
☐ Erase inside	☒ Free-form	☒ Unlimited
☒ Erase outside	☐ Circle	☐ Positive
☐ Fill inside	☐ Rectangle	☐ Negative
☐ Fill outside	☐ Centered	☐ Symmetric

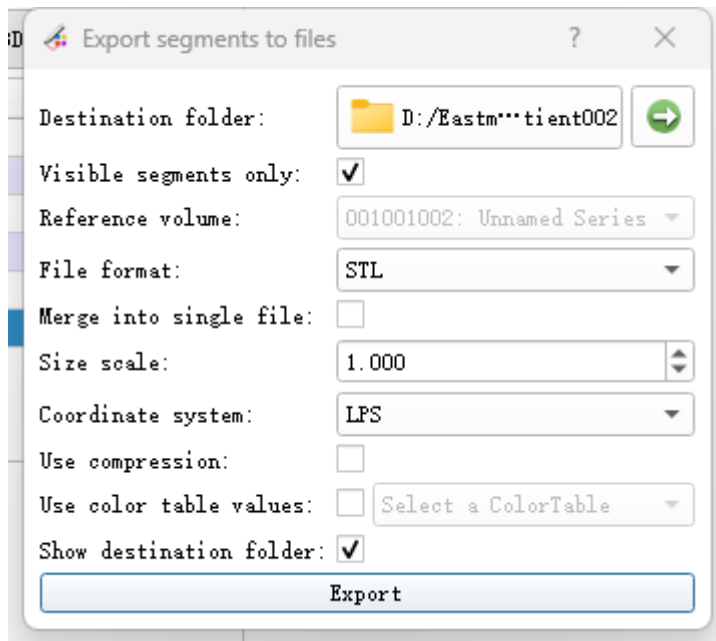
Apply to visible segments: ☐

▼ Masking

Select “erase outside” if you want.



Select the last button and select “Export to files”



Better choose the coordinate system as LPS because it will be in the same coordinate with my program which we will introduce later.

Programme:

There are detailed comments in every line in the code.

All we need to change are the line 8, 90 and 92

8 is the input file path and 90 and 92 are the output file path. 90 is for the upper tooth and 92 is for the lower tooth.

```
Is this an upper tooth or lower tooth? (Enter 'upper' or 'lower'): 
```

When you run the program, it will ask you whether your input file is a upper tooth or a lower tooth. Just enter “upper” or “lower” and then press “ENTER” !!!!

Thank you!! If you have any questions, please contact me by email!

And here is the github of DentalSegmentator:

[GitHub - gaudot/SlicerDentalSegmentator: 3D Slicer extension for fully-automatic segmentation of CT and CBCT dental volumes.](#)

There is more detailed instruction of using this module.